

## **Unnamed\_Unit**

### **Unit 1 - Exploring the Core**

#### **PDF Documents**

##### **PDF Document 1: 1738153357661-Chapter 9 - Unit 1 - Exploring the Core.pdf**

This PDF document is included in the unit materials.



## UNIT 1 EXPLORING THE CORE

### Topic Investigation



#### THE IMPORTANCE OF EFFICIENT WATERING SYSTEMS

Efficient watering systems play a crucial role in maintaining the health of plants and conserving water resources. In areas where water scarcity is an issue, smart watering systems can help reduce wastage and ensure that plants get just the right amount of water they need to thrive.

#### HOW SMART WATERING SYSTEMS CONTRIBUTE TO SUSTAINABILITY

- Smart watering systems use advanced technology to monitor soil moisture and adjust watering schedules automatically. This not only saves water but also promotes healthier plant growth by preventing overwatering or underwatering.
- By reducing water waste, these systems contribute to the sustainability of our environment and help in the conservation of precious water resources.





## Common Problems with Traditional Watering Methods

Traditional watering methods, such as manual watering with a hose or sprinkler systems without sensors, often lead to issues like overwatering and underwatering. Overwatering can cause waterlogging, root rot, and the spread of plant diseases, while underwatering can stress plants and limit their growth.

### OVERWATERING

- Excess water can drown plant roots and lead to fungal diseases.



Too Much!



### UNDERWATERING

- Insufficient water can stress plants and stunt their growth.

### UNEVEN WATER DISTRIBUTION

- Traditional methods may not distribute water evenly across all plants.





## Questioning the Details



### RECENT FLOOD EVENTS

**Q1** How do you think a smart watering system could benefit a garden or farm in your community?



### SUCCESSFUL PREVENTION EFFORTS

**Q2** What are the potential challenges in implementing smart watering technology in areas with limited access to technology?

## Inquiry & Reflection

---

### Discussion question 1

Consider how traditional methods of watering might be adapted to include smart technology. What are the potential benefits and drawbacks of such an adaptation?

### Discussion question 2

Reflect on the importance of water conservation in your community. How can technology help address local water challenges? What role can smart watering systems play in ensuring sustainable water use in both urban and rural settings?